Shared Seas, Competing Interests

Stakeholder Perspectives on Co-Existence Between Fisheries and Offshore Wind Energy in the Southern Skagerrak



Image: AI-generated (prompt) by Dušan Petrović.

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Abstract

Shared Seas, Competing Interests - Stakeholder Perspectives on Co-Existence Between Fisheries and Offshore Wind in the Southern Skagerrak

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This study examines how stakeholders in LivingLab West within the Co-Creating Better Blue programme perceive challenges and opportunities for co-existence between fisheries and offshore wind energy in the southern Skagerrak. It also explores how the LivingLab process supports dialogue and shared understanding. The study is based on nine semi-structured interviews with representatives from the fisheries, wind, and public sectors, analysed through thematic analysis. The findings show that conflict extends beyond spatial competition to include governance, knowledge, and influence. The fisheries sector views offshore wind expansion as displacement and criticises late involvement, while the wind sector highlights fragmented governance and unclear permitting. Governance actors emphasise weak coordination as a key issue. Co-existence is seen as conditional, depending on factors such as fishing methods, wind farm design, and location. Differences in knowledge systems also shape the conflict, as scientific and local knowledge are not equally recognised. LivingLabs can support dialogue, but their legitimacy depends on timing and links to decision-making. The study concludes that sustainable co-existence requires more than technical solutions. It demands earlier, integrated, and meaningful participatory governance where stakeholders can influence.

Keywords: *Co-Creating Better Blue (C2B2); Co-Existence; Fisheries; Marine Spatial Planning; Offshore Wind Energy; Participatory Governance.*

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[1] The author of this study is employed at RISE Research Institutes of Sweden. The study was conducted within the framework of the Co-Creating Better Blue (C2B2) research programme, funded by Mistra.

Abbreviations

Abbreviation	Definition
C2B2	Co-Creating Better Blue
LL*(s)*	LivingLab*(s)*
LLW	LivingLab West
MSP*(s)*	Marine Spatial Plan*(s)* / Planning
OWE	Offshore Wind Energy
RISE	Research Institutes of Sweden
RQ*(s)*	Research Question*(s)*
SwAM	Swedish Agency for Marine and Water Management

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1. Introduction

1.1. Background

For generations, the ocean has been a space where fishing has taken place. More specifically, fishing has been a tradition that can be traced back to the early Stone Age. Fishing is an ancient practice that has served not only recreational purposes but also subsistence and commercial ones. Trawling experienced its major breakthrough during the 1930s and has since become the dominant fishing method (Mattsson, 2024).

In the North Sea, where fishing has long been a traditional activity, offshore wind farms have begun to emerge. The world's first offshore wind farm, Vindeby, was established in 1991 off the coast of Lolland, Denmark, marking the beginning of a rapid expansion of offshore wind energy (OWE) (tethys.pnnl.gov, n.d.). To support the transition to fossil-free electricity production, Sweden's government has granted approval for the OWE project Poseidon in the southern Skagerrak, located off the coast of Gothenburg, Sweden (Westerberg, 2024). The growing demand for fossil-free energy, combined with the rapid expansion of OWE, highlights increasing conflicts of interest among shipping, commercial fishing, environmental protection and renewable energy production (RISE, 2023b). This raises an important question: how can these actors be brought together to engage in dialogue and ensure that their voices are heard in this debate?

Co-Creating Better Blue (C2B2) is a research programme funded by Mistra that works towards a more sustainable, open and democratic blue economy. The programme seeks to create transformative changes by strengthening participatory governance, supported by relevant data and knowledge, and by rethinking how academia, industry, government, civil society and the natural environment interact in decision-making processes. C2B2 also aims to *"give nature a voice"* and to bring stakeholders together around a shared vision for the blue economy and the use of the ocean (RISE, 2023a).

C2B2 is structured around three LivingLabs (LLs) that serve as real-life spaces for collaboration and learning. These LLs bring together actors from different sectors and governance levels who share an interest in marine environments and marine spatial planning (MSP). Each LL focuses on a specific geographic area, selected based on the interests and priorities of the participating stakeholders (Mistra C2B2 Co-Creating Better Blue, 2023).

Within the LLs, stakeholders participate in a series of workshops where they discuss challenges, share perspectives and jointly decide on issues they wish to explore further. This study focuses on LivingLab West (LLW), located on the Swedish west coast, where the southern Skagerrak constitutes the main area of interest.

The aim of this study is to identify how stakeholders within LLW perceive obstacles and solutions related to co-existence between fisheries and OWE. This aim is addressed through a qualitative research approach using semi-structured interviews conducted within the C2B2 programme at Research Institutes of Sweden (RISE). The interviews examine how stakeholders experience conflicts, propose technical, spatial, or dialogic pathways for solutions and reflect on the participatory process itself. The author of this report is employed by RISE, but conducted the study as a master's student within the organisation. Since the author had prior knowledge of the context, participants, and process, this is acknowledged here in the interest of transparency.

1.2. Research Problem

The rapid expansion of OWE in Sweden is transforming marine environments that have long supported traditional uses such as fisheries. In areas like the southern Skagerrak, where fishing has historical, cultural and economic roots, OWE introduces new pressures to already contested marine spaces (Sveriges Fiskares Producentorganisation, 2024). While OWE is central to Sweden's transition towards fossil-free electricity, it also intensifies conflicts between fisheries, renewable energy production, environmental protection and other marine users (Westerberg, 2024; RISE, 2023b).

In the Swedish context, MSP and permitting processes involve multiple authorities and a wide range of stakeholder interests. However, research indicates that invited participation in these processes does not primarily enhance stakeholders' ability to articulate their interests. Instead, it often strengthens planners' capacity to implement policy objectives, while stakeholder participation in the planning process remains vague and weak (Hertting and Hellquist, 2024). This study addresses these gaps by examining stakeholder perceptions of conflicts, barriers and potential solutions for co-existence between fisheries and OWE in the southern Skagerrak. It also analyses how the LL process shapes dialogue, learning and collaborative problem-solving. By focusing on stakeholder experiences rather than abstract

governance models, the study provides context-specific insights that are currently missing from discussions on sustainable marine energy transitions.

1.3. Research Objective

The objective of this study is to explore and analyse stakeholder perspectives on co-existence between fisheries and OWE development in the southern Skagerrak. This study is operationalised through three research questions (RQs). First, the study asks what conflicts and barriers the various stakeholders identify in relation to co-existence. Second, it asks what conditions and pathways are perceived as necessary for enabling co-existence between fisheries and OWE. Third, it examines how the LL process within the C2B2 programme supports learning, participation, and the development of shared understanding among the involved actors. By centering stakeholder perspectives, the study provides empirically grounded insights into how participatory approaches can contribute to managing conflicts and supporting sustainable MSP in practice.

1.4. Previous Research

Co-creation and co-existence between OWE and fisheries has been explored in several European and Nordic studies. As OWE increases in the North Sea and Baltic Sea regions, research has focused on governance mechanisms, MSP, stakeholder engagement, and technological adaptations that enable both sectors to share marine space. A policy brief (2025:28) by Nordregio highlights the growing need to balance renewable energy expansion with traditional marine livelihoods. The report identifies key governance challenges such as fragmented decision-making processes, competing marine space demands, and limited coordination between sectors. It proposes four building blocks for sustainable co-existence: *structured dialogue between stakeholders, participatory planning processes, improved knowledge sharing,* and *supportive institutional frameworks* (Chilvers et al., 2025; pp. 3-4).

When examining the Swedish case, researchers found that collaboration between OWE developers and fisheries has been challenging due to differences between marine spatial planning and project-based permitting systems. While MSP considers environmental, social, and economic impacts at a broader scale, OWE approvals have often relied on project-specific environmental assessments that overlook cumulative effects. Sweden has therefore used a tool called Symphony, developed by the Swedish Agency for Marine and

Water Management (SwAM), to analyse cumulative ecological, social, and economic impacts in marine spatial planning (Chilvers et al., 2025; pp. 13).

A study by Pettersen et al., (2024; pp. 1, 3-4) introduced the MARCO (MARine CO-existence scenario building) concept, which uses scenario modelling and spatial analysis to support stakeholder engagement in OWE planning. The framework combines geographic information systems (GIS) with participatory workshops to explore trade-offs between OWE, fisheries, and environmental protection. By modelling different spatial planning scenarios, the approach helps stakeholders develop a shared understanding of risks, opportunities, and potential compromises in marine space allocation.

Several studies further highlight the role of MSP in managing conflicts between OWE and fisheries. Research on marine co-existence in the Baltic and North Seas shows that interactions between sectors such as OWE, shipping, and fisheries depend strongly on governance structures and national planning systems. Even within the European Union (EU), countries differ in how decision-making processes, licensing frameworks, and stakeholder consultations are organised, which can influence how effectively co-existence between sectors is implemented (Neimane, Eskeland Schütz and Gipperth, 2025; pp. 554-5, 557, 559).

An ongoing project called WIND4COCO, funded by Naturvårdsverket, investigates how offshore wind farms interact with marine ecosystems and fisheries. The project studies biodiversity changes and fish populations around existing offshore wind farms such as Lillgrund in the Öresund strait. It also explores whether adapted fishing gear, such as cagefishing, and monitoring methods could enable fishing activities within or near offshore wind farm areas (Naturvardsverket.se, 2025). Bonsu et al., (2024) had similar findings suggesting that passive fishing methods, such as pot and trap fisheries, may be compatible with offshore wind farm areas, although regulatory gaps and uncertainties about ecological and socio-economic impacts still hinder implementation.

Research by Jiska de Groot et al., (2014) examined interactions between fisheries and marine renewable energy developments in the United Kingdom and identified key challenges such as fishing effort displacement, data limitations, and limited stakeholder communication. The authors argue that mitigation strategies and more collaborative planning processes are necessary to support co-existence between the sectors. Building on this, Szostek et al., (2025)

analysed fishers' perspectives on OWE and found that many fishing communities perceive offshore wind farms as a threat to traditional fishing grounds. However, their study also highlights that improved governance mechanisms, stakeholder engagement, and mitigation measures, such as including fishers as formal consultees in offshore wind planning, can help facilitate co-existence between OWE and fisheries. (Szostek et al., 2025; pp. 9).

2. Methodology

2.1. Empirical Context: The Co-Creating Better Blue LivingLab

This study focuses on stakeholder perceptions of co-existence between fisheries and OWE in the southern Skagerrak. The study is approached through the inclusion of LLs within the C2B2 programme, which provides real-life collaborative settings where stakeholders engage in dialogue, share perspectives, and co-create solutions. In particular, the study draws on LLW as the empirical context. C2B2 consists of five work packages. Work package 4, titled "*the co-creation of participatory ocean governance in LLs"*, focuses on establishing and operating LLs. Three LLs are included in this work. One is located on the Swedish west coast in southern Skagerrak and is referred to as LLW. The other two are situated in the Gulf of Bothnia (LL North) and in the Baltic Sea (LL East) (Wehn, Imperiale and Crystal, 2024; pp. 14).

Within the LLs, workshops were conducted with the aim of developing a clear shared vision while addressing a central challenge formulated by participants in LLW during the early workshop sessions. The identified challenge states that the marine environment in southern Skagerrak is in poor condition according to the Marine Environment Ordinance, which threatens biodiversity and ecosystem functioning as the foundation of the blue economy. At the same time, competition among maritime industries for both resources and space is increasing, creating economic conflicts of interest. Clear guidelines for balancing these conflicts are lacking, as is sufficient political commitment to address them (C2B2 workshop, personal observations, 2025)[2]. The shared vision formulated within the C2B2 LLW is to maintain existing maritime activities and support the development of new ones in collaboration with current and future stakeholders, while also contributing to the restoration,

[2] C2B2 workshop (2025). Personal observation, Gothenburg, 2025-09-10.

improvement, and reduced impact on marine ecosystems (C2B2 workshop, personal observations, 2025)[3].

The guiding principles for the C2B2 LLs emphasizes creating value by understanding stakeholder needs and motivations, enabling stakeholders to influence decisions that affect them, promoting long term economic, social, and environmental sustainability, ensuring openness through broad collaboration and multiple perspectives, and conducting activities in real life contexts (Wehn, Imperiale and Crystal, 2024; pp. 38).

To capture the full innovation potential of LLs and to develop a comprehensive vision for participatory ocean governance in applied real world settings, various stakeholders are involved in the co-creation process. These include representatives from the private sector such as OWE companies, fisheries, and technology companies, as well as public sector, decision makers, implementing authorities, researchers and academics, and civil society organizations (Wehn, Imperiale and Crystal, 2024; pp. 38).

The implementation of the C2B2 LLs follows five interconnected stages: *Prepare, Form, Explore, Embed,* and *Evaluate & Sustain.* In the *Prepare stage*, the organizational team is established, the context and core problems are rapidly analyzed, an initial timeline is created, stakeholder mapping is expanded, and suitable methods are adapted to the local setting. During the *Form stage*, the LL is formally launched, additional stakeholders are invited, shared principles for collaboration are agreed upon, a common vision and collective goals are developed, and key knowledge or data gaps are identified (Wehn, Imperiale and Crystal, 2024; pp. 40-1).

The *Explore stage* focuses on negotiation and the creation of a shared social learning agenda, including clarification of roles, responsibilities, and rules. Actions and data driven innovations are co-created through iterative processes, collaboration among stakeholders at similar governance levels is strengthened both within and across LLs, and participants build capacity to apply relevant assessment and simulation tools where needed. In the *Embed stage*, collaboration continues across governance levels, and the outcomes of social learning are

[3] C2B2 workshop (2025). Personal observation, Gothenburg, 2025-09-10.

integrated both upward into broader governance structures and downward through repeated learning processes among participating actors (Wehn, Imperiale and Crystal, 2024; pp. 40-1).

Finally, the *Evaluate and Sustain stage* involves joint evaluation of results, identification of lessons learned, decisions regarding the future continuation of the LL, and efforts to embed co-creation practices into long term routines that support participatory ocean governance and a sustainable blue economy in Sweden (Wehn, Imperiale and Crystal, 2024; pp. 40-1).

At the time of writing, LLW is in the *Explore stage*, moving toward the *Embed stage*. This progression is not strictly linear, and some stages may be revisited in iterative cycles. This section concludes with a quote from Malmberg et al., (2017; pp. 37), who describes the concept as follows:

"The Living Lab, be it defined as a methodology, ecosystem or community, is foremost about bringing people together to innovate."

2.2. Research Design

This study adopts a qualitative research approach based on individual semi-structured interviews. A qualitative method was chosen as the research aims to explore perceptions, experiences and stakeholder perspectives regarding interactions, co-existence and conflicts between OWE, fisheries and governance in the marine environment. Semi-structured interviews allow for comparability across interviews. They also allow flexibility to follow up on topics raised by the participants. This supports an in-depth understanding of participants' perspectives and experiences. It is particularly well suited to participatory research contexts where diverse stakeholder views are central to the analysis. It enables a deeper understanding of complex and potentially sensitive issues (Adeoye-Olatunde and Olenik, 2021; pp. 1360).

2.3. Researcher Positionality

This study was carried out within the C2B2 programme. Although the author of this study is employed by RISE, the study was conducted in the capacity of a master's student within the organisation. The author had prior knowledge of the organisational context, the participants, and the process under research, and was also involved in observing workshops within the programme. Acknowledging this position is important for transparency, as the author's pre-understanding may have influenced both data collection and interpretation. Within the

LLs, the author's role was to maintain a neutral position and facilitate the workshops. To reduce potential bias, all participants received the same background information prior to the interviews and were interviewed using the same interview guide, with minor adaptations to sector-specific questions. The analysis focused on identifying both similarities and differences across stakeholder groups.

2.4. Sampling & Participant Selection

The empirical material consists of nine semi-structured interviews with representatives from three different sectors: fisheries, wind companies, and public sector. Three participants were interviewed from each sector. At the request of participants from the fisheries sector, one of the interviews was conducted as a group interview with two participants present. Participants were selected through purposive sampling, based on their active participation in LLW workshops and their prior expression of willingness to take part in interviews.

One of the municipalities included in the study did not share the same level of project understanding as the other interviewees. This municipality was included after being mentioned by other participants during the interviews and had previously expressed interest in the C2B2 research programme. Its inclusion was intended to broaden the range of perspectives and strengthen the analytical basis of the study. It should be noted that not all interviewees had the same level of familiarity with the project, as some joined at a later stage. However, all participants were provided with the same preparatory materials and had access to the same background documentation prior to the interviews.

The selected participants are considered knowledgeable actors within their respective sectors, allowing the study to generate insights into broader patterns of stakeholder perspectives and sectoral dynamics related to marine co-existence and governance. Nature protection organisations were not included among the interview participants, as they were not represented in the stakeholder landscape of LLW at the time of the study. The study instead focused on the interaction and dialogue between the fisheries sector, wind sector, and public sector

2.5. Data Collection

Interviews were conducted as one-to-one sessions during February and March 2026 to create a safe and comfortable environment that encouraged participants to share their experiences

and perspectives openly. At the request of participants from the fisheries sector, one of the interviews was conducted as a group interview with two participants present. This approach was considered important given the potentially sensitive nature of sectoral conflicts and governance challenges. The interviews were conducted in Swedish, which is the participants' native language, to minimise misunderstandings and allow for more nuanced responses.

In total, interviews were conducted with three stakeholder groups: the fisheries sector (F1-F3), the wind sector (W1-W3), and public sectors such as planners and municipalities (PS1-PS3). The fisheries sector included two individual interviews and one group interview, where participants are distinguished as F3.1 and F3.2. The wind sector and public sector each consisted of three individual interviews.

Prior to the interviews, all participants received written information about the study and completed a General Data Protection Regulation (GDPR) consent form. The form outlined how personal data would be collected, stored, processed, and clarified the voluntary nature of participation, including the right to withdraw at any time. Interviews were held digitally via Microsoft Teams, were video-recorded and automatically transcribed with the participants' informed consent. During the interviews, the researcher took notes on key arguments and themes raised by the participants. These notes were later used to support the transcription and analysis process.

The interview guide was structured into three thematic blocks aligned with the RQs and these themes also guided the analysis:

- RQ1: Spatial conflicts and barriers.
- RQ2: Conditions and pathways for co-existence.
- RQ3: Dialogue processes and the role of LLs.

The full interview guide, including both Swedish and English versions, is presented in Appendix A. The guide was used as a flexible framework rather than a fixed questionnaire, allowing for follow-up questions and adjustments based on participants' responses during the interviews and sectorial belonging.

Although the author observed several LLW workshops, these observations were not included as empirical material. Workshop documentation and related project outputs were not publicly

available at the time of writing. The observations nevertheless contributed to the author's contextual understanding of the study, while the empirical analysis is based solely on interview data.

2.6. Transcription & Data Handling

The interviews were transcribed using the AI-based transcription service provided by Microsoft Teams. To ensure accuracy and data quality, all transcripts were manually reviewed by listening to the recordings and comparing them with the generated transcripts. Any inaccuracies or missing information were corrected during this process. The combination of AI-based transcription and manual review contributed to both efficiency and reliability. The analysis was conducted in Swedish in order to avoid loss of meaning through early translation. Once the analysis was completed, the analysis was translated into English with the assistance of Microsoft Copilot. AI was also used for overall language support, including grammar and spelling. All analysis, interpretations, discussions, and conclusions were developed by the author.

2.7. Data Analysis

The interview material was analysed using a thematic analysis. Thematic analysis was chosen for this study due to its flexibility and accessibility as a qualitative analytical method. It is a relatively straightforward approach to learn and apply, which makes it particularly suitable for participatory research where participants are engaged as collaborators rather than solely as subjects. Thematic analysis is effective for summarising large qualitative datasets, identifying similarities and differences across perspectives and generating analytical insights. Through the identification and analysis of recurring patterns (themes) in the data, thematic analysis provides a structured way to explore and understand stakeholders' experiences and perspectives (Braun & Clarke 2006; p. 79, 83, 97). Thematic analysis typically proceeds through six interconnected phases:

1) *Familiarisation* with the data through repeated reading and where necessary, transcription.
2) *Systematic generation* of initial codes across the dataset.
3) *Organization* of these codes into potential themes.
4) *Review and refinement* of themes in relation to both coded extracts and the full dataset.

5) *Clear definition and naming* of each theme in relation to the RQ and finally,
6) *The production of a coherent analytical report* that integrates illustrative data extracts with interpretive discussion (Braun & Clarke 2006; pp. 87).

The analysis followed an iterative process in which the transcripts were read multiple times to identify recurring themes, similarities and differences across sectors. Initial codes were generated based on the interview guide and themes emerging from the data. These codes were subsequently refined and grouped into broader analytical categories related to co-existence, co-governance, conflict resolution and transdisciplinary collaboration. This recursive movement between reading, coding, theme development and interpretation reflects the non-linear nature of thematic analysis, in which writing, reflection and refinement occur throughout rather than only at the final reporting stage (Braun & Clarke 2006; pp. 86).

2.8. Literature Review & Data Sources

The literature used in this study was identified through systematic searches in Google Scholar and SöderScholar[4]. These databases were selected as they provide access to peer-reviewed academic literature as well as relevant reports within the fields of environmental governance, MSP, sustainability science and transdisciplinary research.

Searches were conducted using combinations of keywords related to the core themes of the study, including *co-existence; co-creation; co-governance; transdisciplinary; conflict resolution; fisheries* and *offshore wind.* Additional keywords such as *marine spatial planning; maritime spatial planning; stakeholder engagement* and *land-sea interactions* were used when relevant. Reference lists of key publications were also reviewed to identify additional relevant literature. The literature review provided both theoretical framing and contextual background for the study, informed the development of the interview guide and supported the interpretation of empirical findings.

2.9. Ethical Considerations & Challenges

This study addresses ethically sensitive issues related to marine resource use, stakeholder conflicts and participatory governance in the context of OWE and fisheries in the southern

[4] SöderScholar is Södertörn University's library search tool, offering access to e-books, scientific articles, journals, and research publications by searching a broad range of databases, including *Scopus* and *Web of Science.*

Skagerrak. Informed consent and voluntary participation are central, as interviews involve stakeholders from fisheries, wind energy, municipalities and governmental authorities who may hold unequal power positions. Participants are fully informed about the study's purpose, data use and their right to withdraw or skip questions, which is especially important in the LL context where research and collaboration overlap.

Confidentiality and anonymity are also critical, as individuals may be identifiable due to the small stakeholder landscape within LLW and their roles. Data are anonymized, quotations carefully selected and recordings securely stored. The study pays particular attention to power asymmetries, ensuring interviews are respectful, non-confrontational and framed to encourage reflection rather than blame.

Researcher positionality and reflexivity are considered, as conducting the study within the C2B2 programme places the author close to an ongoing participatory process. Transparency regarding potential biases, methodological choices and data interpretation helps maintain ethical and analytical integrity, while ensuring participants' perspectives are represented fairly. Finally, results are disseminated responsibly to avoid conflicts or misrepresenting stakeholder positions. By addressing consent, confidentiality, power dynamics and researcher influence, this study ensures ethical research practices and respects the stakeholders whose experiences underpin the research.

3. Theoretical Framework

This chapter introduces the theoretical framework used in this study. Because the study touches on a broad field of concepts related to co-existence, participation, power, and knowledge, the framework helps create a clearer conceptual understanding of the research problem. The perspectives presented below are used to interpret empirical material and to place the study in relation to wider theoretical discussions. The chapter therefore provides a background for the analysis that follows and helps guide the reader through the central ideas of the study.

3.1. Conceptual & Institutional Framework

3.1.1. Co-Existence & Co-Creation

Co-existence and co-creation can be understood as two distinct yet overlapping concepts within environmental governance. While they refer to different processes and outcomes, both address how multiple actors and users interact, negotiate, and shape shared spaces and resources. When considering the definitions of the terms, co-existence means *"the fact of living or existing together at the same time or in the same place"* and co-creation means *"to make or invent something new together with one or more other people"*, according to the Cambridge Dictionary. These definitions indicate that the two concepts are distinct yet closely related. Co-existence refers broadly to the simultaneous presence of different actors, uses, or interests within a shared space, whereas co-creation emphasizes joint action and the production of new outcomes. In this sense, co-existence can be understood as primarily spatial, as it concerns how different actors occupy and organize shared space. Understanding how these dynamics interact raises further questions about the conditions under which co-existence and co-creation become possible, as well as the factors that may enable or hinder their realization in practice.

Co-existence is commonly understood as a broad conceptual term that describes a condition of multi-use which may take either positive or negative forms. In a neutral sense, it refers to the absence of conflict, where different parties share a resource or interest peacefully, or where any emerging antagonistic interaction is actively managed and resolved (Gee and Mikkelsen, 2023; pp. 28, 36). Co-existence can range from simple co-location, where activities occur in the same space at the same time without shared infrastructure or harmful interference, to more integrated forms of multi-use (Gee and Mikkelsen, 2023; pp. 18).

Beyond spatial considerations, an important internal driver of synergetic sea use is the expectation of improved efficiency and mutual benefit. This occurs when the overall gains from combining activities are greater than the sum of their separate outcomes. External societal drivers, such as strong public support for renewable energy or growing patterns of responsible consumption, can further encourage use and promote co-existence. Increasing spatial pressure and policy developments, including the objective of protecting 30% of marine areas by 2030, also function as external drivers that push sectors toward cooperation and at least neutral co-existence. Economic efficiency is especially significant for emerging sectors and for declining industries seeking diversification as a strategy for survival, which encourages the search for collaborative opportunities. Regulatory conditions, such as licensing systems that support co-use, together with technological innovation that enables new forms of co-existence, for example the expansion of offshore multi-purpose infrastructure, also play an important role (Gee and Mikkelsen, 2023; pp. 19).

A central challenge is the lack of detailed knowledge among planners and decision-makers, not only in relation to spatial data but also regarding the everyday practices and routines of specific users. Synergy is therefore often shaped less by broad sector categories and more by the concrete ways in which users occupy marine space across time and location. Structural barriers may stem from governmental organization, including weak coordination between administrative bodies, inadequate legislation, or licensing regimes that do not encourage co-location. New sectors frequently lack sufficient system knowledge, while industry related challenges can emerge when smaller actors fear they cannot compete with larger and more influential stakeholders. Economic constraints are also critical. When synergies fail to generate direct financial returns, the social and environmental value of cooperation must be clearly demonstrated and supported through policy or incentives (Gee and Mikkelsen, 2023; pp. 20).

However, co-existence alone does not ensure collaboration or mutual benefit. These outcomes often rely on co-creation processes, which introduce a social dimension to how shared marine spaces are governed. Inclusive co-creation is vital for the success and legitimacy, yet it is consistently hampered by several barriers (Davis et al., 2025; pp. 10, 12). Co-creation can be understood as a mainly social process, as it relies on interaction, collaboration, and the integration of diverse perspectives to produce shared outcomes. These include structural barriers like technical jargon and language gaps, systemic drivers such as

power asymmetries and a lack of trust, practical constraints, and design-related barriers like rigid project timelines and tokenistic engagement. These factors often sideline marginalized communities, thereby risking the perpetuation of existing inequalities and undermining project outcomes (Davis et al., 2025; pp. 12, 19-20).

However, the drivers for prioritizing inclusion are strong. Meaningful engagement of marginalized communities leads to richer, more innovative solutions, strengthens community trust and fosters local ownership (Davis et al., 2025; pp. 12, 17). The process requires intentional strategies, including flexible process design, compensating participation costs, valuing all knowledge equally, and using diverse methods to transform governance practices and achieve equitable outcomes (Davis et al., 2025; pp. 33). In this context, MSP may provide an institutional framework capable of facilitating and moderating the barriers to both co-existence and co-creation.

3.1.2. Marine Spatial Planning in Sweden

MSP is established through an EU directive, and in Sweden, MSP is developed through an open and iterative process led by SwAM. The work is conducted in cooperation with national authorities, county administrative boards, municipalities, regions, researchers, industry representatives, and neighbouring countries. The process begins with the collection and analysis of data on current conditions, creating a shared knowledge base for subsequent planning decisions. Draft plans are then produced and subject to two formal consultation rounds, during which authorities, stakeholders, and the public are invited to submit comments (Havs-och vattenmyndigheten, 2025; pp. 6-7). The proposals are revised in response to these inputs before being adopted by the government. The planning framework is grounded in the ecosystem approach, placing ecological limits, biodiversity, and sustainable use at the centre. A strategic environmental assessment evaluates environmental, social, and economic impacts, ensuring that trade-offs between competing interests are made explicit. After adoption, the plans are monitored and revised when necessary, or at least every eight years, forming a continuous cyclical process (Havs-och vattenmyndigheten, 2025; pp. 8-10).

The overall objective of Swedish MSP is to establish long-term and sustainable conditions for sea use up to 2050. By integrating environmental, social, and economic objectives, the plans seek to secure good environmental status, safeguard biodiversity, and promote a sustainable blue economy. The sea is envisioned as a space for competitive and innovative maritime

industries, while natural and cultural values are preserved, climate impacts reduced, and future needs, such as energy production, mineral extraction, and aquaculture, are anticipated (Havs-och vattenmyndigheten, 2025; pp. 10-15). One could argue that MSP is seen as a tool to balance protection and production and to provide guidance, rather than as a legally binding instrument.

Within this framework, co-existence is presented as a guiding principle for managing competing sea uses. Co-existence is described as a flexible arrangement based on mutual adaptation rather than a guaranteed outcome. Co-existence may require regulatory measures or specific permit conditions and is assessed case by case. Compatibility depends on spatial and temporal context, and where uses are considered incompatible, the interest that is most favorable to society is prioritised (Havs-och vattenmyndigheten, 2025; pp. 37).

In relation to OWE and commercial fishing, co-existence is portrayed as particularly challenging by SwAM. Offshore wind farms restrict navigation and fishing activities. Floating offshore wind farms are assessed as incompatible with commercial fishing, and pelagic fishing is considered unable to co-exist with fixed-bottom foundations. Although technical adaptations in offshore wind farm design or fishing practices are mentioned as possible solutions, international experience indicates reduced fishing activity even within adapted wind parks. Further legal and technical development is therefore considered necessary to facilitate co-existence (Havs-och vattenmyndigheten, 2025; pp. 41).

In the North Sea, competition for marine space is especially pronounced, involving fisheries, shipping, and energy production (Havs-och vattenmyndigheten, 2025; pp. 114). Commercial fishing is economically and geographically significant, and cumulative impacts have been considered in the designation of energy areas. Where co-existence with energy extraction is assessed as unfeasible, OWE is designated as the most suitable use (Havs-och vattenmyndigheten, 2025; pp. 115). The plans underline that co-existence requires adaptation (Havs-och vattenmyndigheten, 2025; pp. 119). This brings attention to how such adaptations are negotiated and who is given influence in the process, an issue central to participatory governance.

The open-door system for OWE is a developer-driven model where private companies identify locations themselves and apply for permits to build OWE projects. In Sweden, this

permit-based approach has been seen as beneficial because it encourages innovation, allows a variety of projects, and keeps entry barriers relatively low for developers (Regeringskansliet, 2024, pp. 174). However, this approach is questioned due to limited planning control, given that MSP is not legally binding. This makes it difficult for the state to manage cumulative environmental impacts and to protect sensitive national defence interests. The system also creates uncertainty for energy and grid planning, since developers can obtain permits without being required to build the projects, which can lead to inefficiencies and unnecessary work for authorities (Regeringskansliet, 2024, pp. 71-72).

As a result, a transition to a tender-based system has been proposed, especially since countries such as Denmark and Finland have already moved away from open-door systems (Regeringskansliet, 2024, pp. 18). The government's proposal therefore involves phasing out the open-door model and replacing it with a system where the state designates areas for development in advance, similar to the approaches used in Germany and Denmark (Regeringskansliet, 2024, pp. 174).

3.2. Participatory Governance, Power & Learning

3.2.1. Participatory Governance & Degrees of Participation

Participatory governance refers to involving citizens and stakeholders in public decision-making processes, moving away from purely hierarchical, top-down approaches by state actors. In the context of environmental issues, these "new modes of governance" are highly anticipated, as they are believed to lead to legitimate and effective environmental and sustainability policies. The integration of diverse knowledge, enhanced public acceptance and better conflict resolution are among the hypothesized benefits of participatory environmental governance, which often results in more environmentally stringent decisions (Rose et al., 2025; pp. 2).

Despite the mixed evidence, a study conducted by Rose et al., (2025) provides strong support for the effectiveness of participatory governance in addressing environmental problems, both long-term and short-term. Specifically for challenging long-term environmental problems where standard democratic politics are often hindered by a bias toward short-term success, referred to as democratic myopia. Its effectiveness relies critically on the process design, with intensive deliberation proving to be the key factor for successfully tackling long-term issues,

suggesting that reasoned dialogue is more important than the simple representation of interests in these complex contexts (Rose et al., 2025; pp. 2, 18-19).

Wilcox (1994) builds on Sherry Arnstein's (1969) "ladder of participation" to conceptualise different degrees of public involvement in decision-making. Arnstein's original model consists of eight levels, ranging from non-participation to citizen power reflecting increasing levels of influence over outcomes (Wilcox, 1994; pp. 4-5). Wilcox translates this into five practical "stances" of participation: *information, consultation, deciding together, acting together,* and *supporting independent initiatives* (Wilcox, 1994; pp. 10-15). These stances are summarised in Table 1 below. The stances illustrate how control over decision-making can shift from authorities to stakeholders. At the lower levels, participation is limited to one-way communication or feedback without guaranteed influence. In the middle, stakeholders are actively involved in shaping options and decisions. At the higher levels, participation becomes collaborative or even stakeholder-led, where actors take part in both decision-making and implementation, or are supported in pursuing their own initiatives.

Table 1. Levels of participation based on Wilcox (1994; pp. 10-15)

Stance	**Level of Influence**	**Description**	**Role of Stakeholders**
Information	Very low	One-way communication where stakeholders are informed about decisions or plans. No opportunity to influence outcomes.	Passive recipients of information.
Consultation	Low	Stakeholders are asked for their views, but there is no guarantee that their input will influence decisions.	Provide feedback, limited influence.
Deciding together	Moderate	Stakeholders actively contribute to discussions and help shape decisions alongside authorities.	Co-decision-makers.
Acting together	High	Stakeholders collaborate with authorities not only in decision-making but also in implementing actions.	Partners in decision-making and implementation.
Supporting independent initiatives	Very high	Stakeholders are empowered to lead their own initiatives, with authorities providing support such as resources or guidance.	Lead actors with support from authorities.

A key point is that no single level is inherently preferable. Instead, the appropriate stance depends on the context, the purpose of participation, and stakeholder expectations (Wilcox, 1994; pp. 3-5, 8). In addition, Wilcox emphasises participation as a process over time, including phases such as initiation, preparation, participation, and continuation (Wilcox, 1994; pp. 8-9, 16, 18). The framework also highlights the importance of recognising diverse stakeholders with different interests and desired levels of influence. Effective participation therefore requires aligning the level of involvement with both the context and stakeholders' roles in decision-making, as well as being transparent about where decision-making power ultimately lies (Wilcox, 1994; pp. 3, 8-9, 25).

3.2.2. Reflexive Governance & Learning

Reflexive governance emphasizes the need to continuously and critically evaluate governance systems as environmental challenges become more complex and change over time. Reflexive governance emerged from the recognition that earlier policy approaches, including regulation, market-based instruments, and participatory mechanisms, often failed to achieve lasting environmental improvements due to deeper structural constraints within economic and political systems (Feindt and Weiland, 2018; pp. 662). Rather than focusing only on improving policy instruments, reflexive governance emphasizes the transformation of institutional structures, norms, and underlying assumptions that shape decision-making.

This perspective understands governance as a learning-oriented process, where reflection and dialogue create feedback across policy areas and encourage actors to reconsider established practices, knowledge, and values. In this way, reflexive governance complements more traditional forms of coordination by integrating reflection, dialogue, and social learning into governance processes. This is particularly important for addressing long-term and complex sustainability challenges that cannot be managed through linear or sector-specific approaches (Feindt and Weiland, 2018; pp. 663-665).

However, this potential is limited by structural and political conditions, such as institutional inertia, established practices, and power imbalances that influence whose knowledge is recognized and whose perspectives shape outcomes. As a result, reflexive governance does not resolve power issues but operates within them, making it necessary to critically examine how learning, participation, and influence are distributed in practice (Feindt and Weiland, 2018; pp. 667-670).

3.2.3. Power Asymmetries in Marine Governance

Even when participatory processes are formally in place, they do not automatically eliminate power imbalances. Participation often unfolds within existing institutional and economic structures, where differences in authority, resources, and dependency continue to shape who can meaningfully influence outcomes. Power imbalance deeply influences the dynamics between organizational actors, which is particularly relevant when considering the relationship between established industries such as fisheries and emerging sectors like OWE. This imbalance, also referred to as power asymmetry or dominance, arises when a stronger actor can impose unfavorable conditions on weaker partners, undermining the confidence and commitment required for constructive collaboration (Essabbar et al. 2016; pp. 1022-1023).

The implications of this imbalance are most severe in scenarios of "strong domination" or "strong subordination". For example, if the OWE industry achieves a position of strong dominance over the fishing sector, the latter may find itself without viable alternatives, becoming highly vulnerable to the OWE sector's opportunism regarding access, compensation, or mitigation measures. If the fishing sector experiences strong subordination, it faces a "difficult situation" where the primary strategic goal becomes mitigating adverse impacts and restructuring its engagement to shift the power balance. Understanding these power situations is essential for predicting negotiation outcomes and crafting policies that ensure equity in marine resource allocation (Essabbar et al. 2016; pp. 1029-1031).

3.2.4. The Trinity of Voice

In this context, ensuring that participatory and reflexive governance processes translate into meaningful engagement requires attention not only to formal structures but also to the quality of interaction between actors. Senecha's study (2023) introduces the Trinity of Voice framework, which conceptualizes effective participation through three interrelated dimensions: *access, civic standing,* and *influence. Access* refers to the ability of stakeholders to enter and meaningfully engage in participatory processes, *civic standing* concerns being respected, heard, and recognized as legitimate contributors, and *influence* relates to the extent to which participants can shape outcomes. Together, these dimensions highlight that participation is not only about formal inclusion but also about the quality of interaction and the distribution of voice within decision-making processes. Addressing these aspects helps build trust, reduce conflict, and create conditions for more inclusive and legitimate

governance, particularly in complex environmental contexts where competing interests and uncertainties are present (Senecah, 2023; pp. 8-9, 18-19).

3.3. Experience-Based Knowledge vs. Scientific Knowledge

Experience-based knowledge, often referred to as local ecological knowledge, is a professional, very precise form of understanding acquired through practical, lived experience, especially within fields like fishing. This knowledge is primarily learned through intergenerational transfer (from fathers, grandfathers, and uncles to sons) often within a community of practice. Fishers gain this understanding by interacting with the environment, observing factors like habitat, season, and weather, which inform their ecological knowledge, including fish movement and behavior patterns (Ervik and Strømme, 2023; pp. 1-2). This embodied knowledge encompasses practical skills, a professional vocabulary and detailed, intimate, fine-scale ecological information crucial for understanding biodiversity and ecosystem complexities that may not be available otherwise (Ervik and Strømme, 2023; pp. 4). While this knowledge is credible and described as "ancient knowledge" developed in interaction with the environment, its reliability is sometimes questioned because it is nonscientific and may require verification or "translation" before acceptance as scientific knowledge (Ervik and Strømme, 2023; pp. 1).

Scientific knowledge, on the other hand, is formalized, structured information derived from systematic research, such as environmental monitoring. While the scientific community contributes to both research and teaching in science, it is increasingly recognizing the value of citizen science and fishers' experience-based knowledge as supplementary documentation and empirical data. The challenge in integrating these two knowledge systems lies in the fact that local ecological knowledge often does not fit into the established models and governance instrumentation used in Western fisheries and resource management (Ervik and Strømme, 2023; pp. 1-2). Nevertheless, researchers who use fishers' summaries and stories as raw material and systematize their findings can formalize this experience-based knowledge, making it usable as empirical evidence in scientific research (Ervik and Strømme, 2023; pp. 7). This integration is vital for addressing complex and compound environmental issues, such as environmental pollutants in marine species, where multiple data sources are necessary to gain a comprehensive overview (Ervik and Strømme, 2023; pp. 1-2).

4. Results & Analysis

This chapter presents the results of the thematic analysis based on the interview material and the method by Braun and Clarke (2006). To provide a clear overview of the empirical findings, the results are organised according to the three stakeholder groups included in the study: the fisheries sector (F1-F3), the wind energy sector (W1-W3), and public sector (PS1-PS3). Within each section, the findings are presented through the main themes identified in the analysis, showing how the different actors perceive conflicts (RQ1), conditions and pathways for co-existence (RQ2), and the role of dialogue processes such as LLs (RQ3).

4.1. The Fishing Sector's Perspective

The thematic analysis of the interviews with representatives from the fishing sector resulted in four central themes (Table 2). These themes illustrate how actors within the fishing sector perceive conflicts between fisheries and OWE in the southern Skagerrak, the opportunities for co-existence that are identified, and how dialogue processes such as LLs are perceived.

Table 2. Final themes from the thematic analysis and their descriptions in relation to the fishing sector.

Final Themes	Description of the Themes
Theme 1. Spatial displacement and loss of fishing grounds	This theme describes how actors within the fishing sector perceive OWE as a process that displaces fisheries from important fishing grounds. The conflict primarily concerns spatial competition over marine areas, where offshore wind farms and their safety zones restrict access to areas traditionally used for fishing.
Theme 2. Criticism of the planning process and lack of early dialogue	This theme addresses the fishing sector's criticism of how OWE planning is conducted. Respondents perceive that fisheries are often included late in the process, when key decisions have already been made.
Theme 3. Co-existence perceived as limited and dependent on fishing method	This theme describes the fishing sector's view on the potential for co-existence between OWE and fisheries. Respondents do not reject co-existence entirely, but perceive it as highly limited and dependent on fishing methods and location.
Theme 4. LLs are seen as valuable but dependent on timing and actual influence	This theme concerns the fishing sector's perception of dialogue platforms such as LLs. Respondents see value in bringing different actors together, but emphasize that such processes must be linked to actual decision-making processes to be perceived as meaningful.

Theme 1. Spatial displacement and loss of fishing grounds

A central theme in the interviews is the experience of spatial displacement from important fishing areas. Several respondents describe how offshore wind farms overlap with key fishing grounds, particularly for species such as shrimp and Norway lobster. One respondent describes the situation as OWE being placed directly in the most productive and attractive fishing areas:

"They have taken over all the best fishing waters." (F2)

Respondents emphasize that fishing in the area is already geographically limited, and that additional restrictions therefore have significant consequences for their operations. When offshore wind farms are established together with safety zones, access to areas traditionally used for fishing is restricted. This is described as particularly problematic because fishing often depends on specific seabed conditions and the migration patterns of fish. At the same time, the interviews show that perceptions of conflict vary between different types of fishing. One respondent engaged in fishing methods other than bottom trawling suggests that OWE would likely not negatively affect their activities:

"I do not expect any impact at all… I rather believe it could be positive with such areas where there is an automatic protected zone." (F1)

This indicates that the perception of spatial conflict is closely linked to fishing methods and the geographical use of marine areas. The theme shows that the conflict between fisheries and OWE is primarily experienced as a territorial conflict. Fisheries view marine space as a key production resource, and when these areas are claimed by other sectors, a sense of displacement arises. At the same time, the variation between F1 and F2-F3 demonstrates that the conflict is not equally strong across all types of fishing.

Theme 2. Criticism of the planning process and lack of early dialogue

A recurring theme in the interviews is criticism of how OWE planning is conducted. Respondents perceive that the fishing sector is often included late in the process, when key

decisions regarding site selection have already been made. For example, one respondent describes how fisheries are not involved at an early stage when project areas are identified:
"Wind companies have never spoken to the fishing sector before selecting their areas." (F3.1)

Instead, several respondents experience that dialogue mainly takes place within formal consultation processes, where the opportunity to influence project location is already limited. This is perceived as problematic, as many conflicts could have been avoided through early dialogue. One respondent argues that conflicts could have been reduced if fisheries had been involved at an earlier stage of planning:

"If we had been involved from the beginning… there are waters where we do not fish much today that would have been suitable for wind companies to use." (F2)

The results show that criticism of the planning process largely concerns the lack of early dialogue and the limited ability of the fishing sector to influence the location of offshore wind farms. The theme demonstrates that the conflict is not only about physical marine space but also about procedural justice. The fishing sector feels that its knowledge and experience are not sufficiently incorporated at an early stage of the planning process. This contributes to distrust towards both companies and governance systems and reinforces the perception of conflict.

Theme 3. Co-existence perceived as limited and dependent on fishing method

Regarding the potential for co-existence between fisheries and OWE, a more nuanced picture emerges from the interviews. Respondents do not reject co-existence as a concept, but argue that the possibilities are highly dependent on technology, location, and fishing methods. Several respondents emphasize that floating offshore wind farms in practice means that fishing can no longer be carried out in the area. At the same time, skepticism is expressed towards certain solutions often highlighted in the debate, such as aquaculture within offshore wind farms. These ideas are perceived by several respondents as difficult to implement in practice:

"If we are going to have a debate about this, it needs to be a balanced discussion and not about fantasies and propaganda." (F2)

While some actors see potential ecological benefits of OWE, others perceive technical and practical limitations that make co-existence difficult and highly constrained in practice. The analysis therefore indicates that the potential for co-existence largely depends on fishing method, location, and offshore wind farm design.

Theme 4. LLs are seen as valuable but dependent on timing and actual influence

The fourth theme concerns how respondents perceive dialogue platforms such as LLs. Several respondents see value in forums where different stakeholders can meet and discuss shared issues. For example, one respondent describes how such meetings can contribute to increased understanding between actors:

"It is always good to understand the perspectives and starting points of other special interests." (F1)

At the same time, the respondent feels that discussions sometimes become relatively theoretical and distant from the practical reality in which fisheries operate:

"I see it as very theoretical… I mainly operate in the practical everyday reality." (F1)

Timing is also highlighted as a key factor. One respondent argues that dialogue processes would have had greater impact if they had been conducted earlier in the planning process:

"I have to say that the main weakness is the timing. Most applications are already in their formal processes when we have our LLs. There is no function then." (F3.2)

Despite these limitations, participation in such forums appears to be important for bringing forward the perspectives of the fishing sector in discussions on OWE. The results show that the fishing sector is generally positive towards dialogue platforms such as LLs, but that the legitimacy depends on when in the process it occurs and the extent to which it can influence decision-making. When dialogue is perceived as too late in the process or overly theoretical, there is a risk that it is seen as symbolic rather than as a practically relevant tool.

4.2. The Wind Sector's Perspective

The thematic analysis of the interviews with representatives from the wind sector resulted in four central themes (Table 3). These themes illustrate how actors within the wind energy industry perceive conflicts with fisheries, the opportunities for co-existence that are identified, and how dialogue processes and governance systems influence the development of OWE in Sweden.

Table 3. Final themes from the thematic analysis and their descriptions in relation to the wind sector.

Final Themes	Description of the Themes
Theme 1. The conflict is primarily understood as a governance issue, not only as a sectoral conflict	The wind sector does not primarily describe the conflict with fisheries as a straightforward opposition between two industries. Instead, the conflict is understood as being shaped by an unpredictable system, including the open-door model, unclear marine spatial plans, multiple parallel permitting processes, defense interests, municipal politics, and varying assessments by authorities.
Theme 2. Co-existence is seen as possible, but only under certain technical and spatial conditions	Wind energy actors describe co-existence as something that may be possible in certain cases, but not in all. A key distinction is made between fixed-bottom offshore wind farms, where some trawling can continue, and floating offshore wind farms, where bottom trawling in practice ceases. Co-existence is therefore seen as conditional on factors such as technology, design, compensation, distance, depth, cables, and fishing method.
Theme 3. The conflict also concerns different knowledge systems and perceptions of reality	A central finding is that wind actors perceive the conflict as not only concerning space and economic interests, but also what counts as valid knowledge. Research, environmental impact assessments, models, and data are contrasted with experiential knowledge and a lack of trust in institutions by the fisheries sector. Wind actors describe this as one of the main challenges in dialogue.
Theme 4. Dialogue is important, but constrained by timing, distrust, and strategic behavior	The wind sector strongly emphasizes the need for dialogue, but also highlights that it often takes place under poor conditions: too late in the process, after the locations of offshore wind farms have already been decided, while permits are being appealed, or within forums where actors are reluctant to share information fully.

Theme 1. The conflict is primarily understood as a governance issue, not only as a sectoral conflict

A central theme in the interviews is that wind energy actors do not primarily describe the conflict with fisheries as a direct opposition between two industries. Instead, the conflict is understood as a consequence of a complex and partly unpredictable governance system, where multiple actors, regulations, and decisions influence the development of OWE. One respondent describes how developers themselves must identify potential project areas by gradually filtering out locations where other interests are already present:

"We would prefer a system where the state designates areas… but that is not how the Swedish system works. All developers have to try to find locations themselves." (W1)

In this process, project developers must consider a wide range of overlapping interests, such as defense, shipping, environmental values, and commercial fishing. As a result, the areas that remain are often already limited. At the same time, respondents describe the permitting processes as complex and difficult to predict. One respondent characterizes the situation as involving multiple authorities and regulatory frameworks interacting in ways that make the process difficult to oversee. Several respondents also argue that different authorities tend to prioritize their own sectoral interests without weighing them against each other from a broader societal perspective:

"Each authority looks at its own interest… but no one makes the socio-economic trade-off." (W2)

The results show that wind energy actors largely interpret the conflict with fisheries as an issue of systemic governance. A lack of coordination between planning processes, permitting procedures, and sectoral interests contributes to conflicts between different uses of marine space. For the wind sector, the problem is therefore not only resistance from fisheries, but also that the state has not yet established a clear and legitimate framework for how priorities in marine areas should be determined. This clearly relates to how conflicts emerge within planning and governance processes.

Theme 2. Co-existence is seen as possible, but only under certain technical and spatial conditions

A second central theme in the interviews concerns perceptions of co-existence between fisheries and OWE. Respondents do not reject the possibility of co-existence, but emphasize that it is highly dependent on technology, design, and location. An important distinction is made between different types of offshore wind farms. In parks with fixed-bottom foundations, respondents suggest that certain types of fishing can still be carried out. This is enabled by turbines being placed at relatively large distances from each other, allowing fishing vessels to navigate between them. One respondent describes this as a potential form of co-existence:

"If the turbines are spaced far enough apart, it is possible to fish in between them." (W2)

At the same time, it is emphasized that floating offshore wind farms impose greater limitations on fishing, particularly for bottom trawling, as mooring systems and cables create obstacles:

"With floating technology, it effectively results in the exclusion of bottom trawling." (W3)

Respondents also highlight alternative fishing methods, such as pot fishing or other forms of passive fishing, as possible adaptations within or near offshore wind farms. However, these solutions often face resistance from the fishing sector, as they require changes to established fishing methods and practices. Aquaculture is also mentioned as a potential alternative to enable co-existence.

The results thus show that the wind sector considers co-existence to be possible, but strongly conditional. The extent to which different activities can co-exist is seen as depending on factors such as technological solutions, offshore wind farm design, and the fishing methods used. Co-existence therefore does not appear as a simple yes-or-no issue, but rather as a matter of design and adaptation. This relates to which solutions are perceived as realistic for enabling co-existence from the perspective of the wind sector.

Theme 3. The conflict also concerns different knowledge systems and perceptions of reality

A third theme in the analysis concerns differences in how wind energy actors and the fishing sector perceive and interpret knowledge about the marine environment. Several respondents describe that the discussion is not only about economic or spatial competition, but also about what counts as legitimate knowledge. One respondent, for example, explains how research findings and environmental impact assessments are often met with skepticism from the fishing sector:

"They have quite low trust in authorities and research." (W1)

At the same time, wind actors perceive that they and the fishing sector sometimes base their arguments on different types of knowledge. One respondent describes how the parties can arrive at different conclusions despite analyzing the same area:

"We can talk about the same area and the same data, but still reach different conclusions." (W3)

In the interviews, this is linked to a broader distinction between research-based knowledge and the local ecological knowledge developed through practical fishing activities. One respondent explains how fishers often refer to their daily experience at sea and argues that they know their fishing areas better than individuals working in offices or producing scientific studies.

The results indicate that the conflict is partly epistemological, rooted in differing knowledge systems and perceptions of reality. Actors assign varying levels of legitimacy to scientific expertise and practical experience, reflecting a divide between the knowledge bases of the wind and fishing sectors. This divergence shapes how ecological effects, risks, and uncertainties are interpreted, and influences how the conflict itself is understood.

Theme 4. Dialogue is important, but constrained by timing, distrust, and strategic behavior

The fourth theme concerns how wind energy actors perceive dialogue processes and collaborative initiatives. Several respondents emphasize that dialogue with other actors is

essential for enabling co-existence between different uses of marine space. One respondent describes a combination of dialogue, technological development, and compensation as key conditions for finding solutions. At the same time, it is emphasized that dialogue often takes place under challenging conditions. In some cases, dialogue occurs simultaneously with ongoing or appealed permitting processes, creating a situation where actors have different strategic incentives. One respondent describes this as problematic:

"We are supposed to discuss co-existence while the fishing sector is appealing our permits. Does that really show that fisheries want to co-exist?" (W1)

Several respondents also argue that some actors may be reluctant to share information fully, as issues related to location and project development also involve competition between companies within the same sector. Information about project areas or strategic plans may therefore be perceived as commercially sensitive. This can make actors more cautious about what information they share in dialogue and collaboration processes. At the same time, it is noted that meeting in person can help improve relationships and create greater understanding of each other's perspectives:

"There is value in simply being in the same room and talking. It creates an opportunity to understand each other's perspectives, needs, and challenges." (W3)

Overall, the results show that dialogue processes are perceived as important, but that their effectiveness may be limited by factors such as timing, strategic interests, and a lack of trust between actors. The overall picture is relatively consistent: the wind sector sees LLs as valuable for building relationships and increasing mutual understanding, but also calls for clearer objectives, a more concrete focus, and better timing within the process. This directly relates to how actors experience dialogue, participation, and LLs.

4.3. The Public Sector's Perspectives

The thematic analysis of the interviews with representatives from the public sector resulted in four central themes (Table 4). These themes illustrate how public sector actors at the national and municipal levels perceive the conflict between fisheries and the expansion of OWE, the challenges in governing marine areas, and the opportunities they identify for improved coordination and dialogue between actors.

Table 4. Final themes from the thematic analysis and their descriptions in relation to the public sector.

Final Themes	Description of the Themes
Theme 1. The conflict is understood as a spatial conflict exacerbated by unclear governance	Public sector actors describe the conflict between OWE and fisheries as a classic competition over the same marine space, but emphasize that the problem is exacerbated by governance that is not sufficiently clear or coordinated. MSP are guiding rather than binding, and municipalities, authorities, and the state operate at different levels. As a result, uncertainty itself generates further conflict and ambiguity.
Theme 2. Fisheries are perceived as both legitimately vulnerable and difficult to engage with in dialogue	Public sector actors express a clear understanding that the fishing sector perceives itself as under pressure, economically, culturally, and politically. At the same time, they describe the behavior of the fishing sector as forceful, vocal, and sometimes difficult to engage with in constructive dialogue. This leads to fisheries being perceived both as a sector affected by development and as a resistant actor.
Theme 3. Solutions are primarily seen in early, integrated, and multi-level planning	Public sector actors repeatedly emphasize that solutions do not primarily lie in making late-stage adjustments. Instead, they argue for planning processes in which fisheries, energy, environmental concerns, defense, and local perspectives are considered earlier and more clearly. In practice, several call for a more integrated system.
Theme 4. LLs and workshops are seen as valuable, but their usefulness depends on whether they lead to concrete decision relevance	Public sector actors are generally positive towards LLs as a method. They see value in bringing different actors together, facilitating knowledge exchange, and increasing mutual understanding. However, they also stress that the format needs to become more concrete, more continuous, and more clearly linked to actual decision-making processes in order to be effective.

Theme 1. The conflict is understood as a spatial conflict exacerbated by unclear governance

A central theme in the interviews is that the conflict between fisheries and OWE is primarily understood as a competition over the same marine space. At the same time, the public sector actors emphasize that the conflict is reinforced by how the governance of marine areas are organized. One respondent describes how MSP in Sweden function as guiding rather than binding instruments, creating uncertainty about how they should be interpreted in practice. This means that multiple actors (authorities, municipalities, and the state) operate at different levels within the decision-making process. According to respondents, this creates a situation in which actors interpret planning differently, which in itself can intensify conflicts between sectors. One respondent, for example, explains how planning can sometimes result in larger areas being identified than will actually be developed, which may create concern among other sectors:

"There is a need to overplan… perhaps two-thirds of the areas will disappear in these plans at a later stage." (PS1)

From the perspective of fisheries, however, this may be perceived as large parts of the sea being at risk of being claimed, thereby reinforcing the perception of conflict. At the municipal level, the conflict is similarly described primarily as an issue of how marine space should be allocated between different uses. One respondent characterizes this as a classic conflict between two industries:

"The classic conflict is about space." (PS2)

The results thus show that public sector actors largely interpret the conflict between fisheries and OWE as a spatial conflict, which is further intensified by unclear or fragmented governance.

Theme 2. Fisheries are perceived as both legitimately vulnerable and difficult to engage with in dialogue

A second theme concerns how public sector actors perceive the role of fisheries in the conflict. Respondents often express a clear understanding that the fishing sector feels under pressure, both economically and culturally. One respondent, for example, describes how fisheries have, over time, faced increasing regulation and new competing interests at sea:

"They have gone from a kind of unregulated freedom at sea to an extremely complex system." (PS1)

At the same time, fisheries are sometimes perceived as a challenging dialogue partner in conflicts related to OWE. One respondent describes how reactions from the fishing sector can, in some cases, be strong and confrontational, including the use of worst-case scenarios. Similar observations are made at the municipal level, where protests from fishers are described as intense and highly visible in the public debate:

"There are very strong emotions involved." (PS2)

At the same time, there is an understanding of why fisheries react strongly. Respondents emphasize that fishers often perceive their future opportunities to operate as being restricted, and that the cultural value of their livelihood is threatened by sharing marine space. The results therefore show that, from the perspective of public sector actors, fisheries appear both as a legitimately vulnerable sector and as an actor that can sometimes be difficult to engage with in constructive dialogue.

Theme 3. Solutions are primarily seen in early, integrated, and multi-level planning

A third central theme concerns how conflicts between different uses of marine space could be better managed in the future. Several respondents emphasize that problems often arise because planning occurs too late in the process, when projects have already been developed and conflicts have already emerged. One respondent describes how the process in Sweden is largely driven by project developers who independently identify areas for OWE:

"Wind power companies claim their areas without any dialogue with either fisheries or municipalities." (PS3)

Several respondents argue that a more integrated planning system could reduce conflicts. At the municipal level, for example, it is suggested that the state could play a greater role in identifying areas for different uses in advance. Similar reasoning is also expressed at the authority level, where one respondent emphasizes the need to balance different objectives within planning:

"We have goals on the energy side and goals for fisheries. Planning should try to reconcile them." (PS1)

The results indicate that public sector actors largely see the solution to conflicts in earlier, more integrated, and multi-level planning processes, where different sectoral interests are considered and balanced from an early stage.

Theme 4. LLs and workshops are seen as valuable, but their usefulness depends on whether they lead to concrete decision relevance

The fourth theme concerns how public sector actors perceive dialogue platforms such as LLs and joint workshops. Respondents are generally positive towards these formats and see clear value in bringing different actors together to share knowledge. For example, respondents describe how such meeting spaces can contribute to increased understanding between actors:

"They become people you meet, and you begin to understand their perspectives." (PS1)
"It generates knowledge exchange and in-depth dialogue." (PS2)

At the same time, it becomes clear that dialogue platforms only gain real significance if they are linked to actual decision-making processes. One respondent emphasizes that the outcomes of such processes need to contribute to resolving concrete conflicts between actors:

"If the parties were to agree… then it could be brought to the political level with concrete solutions." (PS3)

Several respondents also highlight that dialogue processes risk becoming overly theoretical if they are not connected to practical issues or specific geographic areas. Overall, the results show that LLs are perceived as valuable forums for dialogue and learning, but that their ultimate significance depends on whether they contribute to concrete solutions that can be applied in decision-making processes.

The findings show that the conflict between fisheries and OWE extends beyond competition for marine space and is also shaped by governance structures, knowledge differences, and perceptions of influence. Across all stakeholder groups, early dialogue and more integrated planning were highlighted as important conditions for enabling co-existence. While LLs were generally viewed as valuable platforms for learning and exchange, their effectiveness was considered dependent on timing and their connection to actual decision-making processes.

5. Discussion

This chapter explains what the findings mean by examining why conflicts arise between fisheries, OWE, and governance actors, how co-existence is understood in practice, and the role of collaborative processes like LLs. It then unpacks the underlying dynamics of these conflicts, explores possible pathways for co-existence, and reflects on the opportunities and limitations of current governance processes.

5.1. Understanding the Conflict: Beyond Spatial Competition

At first glance, the conflict between fisheries and OWE in the southern Skagerrak region appears to be a straightforward competition over marine space. However, the results indicate that the conflict is more complex than a simple spatial overlap. While all three actor groups recognise space as a central issue, they interpret the nature and causes of the conflict differently. For the fisheries sector, the conflict is closely tied to displacement from historically important fishing grounds and to the loss of areas that carry not only economic value, but also cultural and professional meaning. Fishers describe these waters as places where fishing has "always" taken place, which suggests that the conflict is rooted in long-standing relationships to place, practice, and identity rather than only present-day resource competition.

The wind sector does not primarily frame the conflict as a clash between two equal sectors competing for the same space. Instead, wind actors tend to interpret conflict as an effect of fragmented governance, unclear priorities, and an unpredictable planning and permitting system. Their perspective shifts attention from the sea itself to the institutional structures through which decisions about the sea are made. Governance actors occupy a more ambivalent position. They largely present themselves as neutral coordinators rather than direct parties to the conflict, even though the results suggest that municipalities and public actors do not always remain fully impartial in practice. Based on the empirical findings, PS2 appears more sympathetic to wind development, while PS3 expresses greater understanding for fisheries' concerns. This illustrates that governance actors are not outside the conflict, but are themselves embedded within it, balancing sectoral interests while also being shaped by local political and institutional priorities.

These contrasting interpretations can be linked to theory on participatory governance, power, and voice. From the perspective of Wilcox (1994), participation is not only about being present in a process, but about the degree of actual influence actors are given. The fisheries sector repeatedly describes being brought in too late, often after key locations have already been selected. This suggests a form of participation closer to consultation than deciding together. As Wilcox argues, increased participation can strengthen legitimacy and mutual understanding, but only when participants are involved early enough to shape outcomes. This also connects to Senecah's Trinity of Voice (2023), particularly the dimensions of access and influence. Fishers may formally have access to consultation processes, but their accounts suggest weaker influence over decisions that directly affect their livelihoods.

At the same time, the results support Feindt and Weiland's (2018) argument that governance processes do not operate outside power asymmetries. Conflict is therefore not something that can realistically disappear altogether. Competing interests in marine space will remain, especially where large-scale energy transitions intersect with traditional livelihoods. Yet the persistence of conflict does not mean participation has failed. A stakeholder not getting its preferred outcome does not automatically mean that it has not been heard. Rather, the key question is whether the process has created fair conditions for voice, recognition, and negotiation despite disagreement.

At the same time, conflict should not only be understood as a problem that always can or should be resolved. As Tafon et al., (2021; pp. 214-215) argue, conflicts can also act as catalysts for transformative change by revealing underlying inequalities and mobilising actors to engage more actively in governance processes. In this sense, conflict may contribute as a 'window of change' towards more inclusive and just forms of ocean governance, rather than simply representing a barrier to co-existence. Recognising conflict as a normal and unavoidable part of social interaction therefore shifts the focus from eliminating disagreements to managing them constructively.

Overall, the findings suggest that the conflict should be understood not only as spatial competition, but as a conflict shaped by history, identity, governance structures, and unequal influence. This answers RQ1 by showing that the barriers to co-existence are simultaneously territorial, procedural, and relational. The next section shifts focus from why tensions arise to what might be done about them, by exploring how co-existence is perceived in practice and under what conditions it may, or may not, be possible.

5.2. Co-Existence in Practice: Conditions, Limits & Trade-Offs

When comparing how the three actor groups perceive solutions for co-existence, it becomes clear that co-existence is not understood as a fixed outcome, but as something highly conditional. Across all sectors, the possibility of co-existence is closely tied to the type of OWE technology, spatial design, and fishing practices involved. The wind sector tends to emphasise technical solutions and spatial design as key enablers. For instance, fixed-bottom offshore wind farms are described as potentially allowing certain types of fishing to continue between turbines, whereas floating offshore wind farms are widely seen as incompatible with bottom trawling due to mooring systems and cables. Governance actors largely support this conditional view, framing co-existence as something that must be addressed through earlier and more integrated planning processes.

In contrast, the fisheries sector expresses a more sceptical position. While co-existence is not rejected, it is often perceived as limited in practice and dependent on significant changes to existing fishing methods. Proposed alternatives such as passive fishing techniques or aquaculture are frequently met with resistance, as they require not only technical adaptation but also shifts in established practices and livelihoods. This highlights an important divergence where wind and governance actors often approach co-existence as a technical and

a planning challenge, and the fisheries sector experiences it as a deeper transformation of how marine space is used and what it represents.

These differences can be understood through the theoretical distinction between scientific and experience-based knowledge (Ervik and Strømme, 2023). Technical solutions for co-existence are often grounded in scientific models, environmental assessments, and spatial planning tools, whereas fishers rely on local ecological knowledge shaped by long-term, embodied interaction with the marine environment. As a result, suggested solutions may appear detached from practical realities or difficult for fishers to implement. This gap between knowledge systems makes it more difficult to translate technical possibilities into socially accepted solutions, thereby reinforcing the tensions identified in the findings.

Wind actors highlight how scientific data and environmental impact assessments are sometimes met with scepticism, while also recognising that both sectors can interpret the same information differently. This suggests that the disagreement is not necessarily about the absence of knowledge, but about competing ways of understanding and valuing it. In this sense, different knowledge systems co-exist but are not easily reconciled within current governance and planning frameworks. Scientific and experience-based knowledge should not necessarily be understood as competing forms of knowledge. Rather, they can be viewed as complementary, where scientific knowledge provides broader evidence-based insights and experience-based knowledge contributes place-based and practice-oriented perspectives. Combining these forms of knowledge may support a more comprehensive understanding of marine environments and the challenges related to co-existence.

At the same time, power asymmetries further shape what forms of co-existence are considered realistic. As discussed by Essabbar et al., (2016), weaker actors may be required to adapt to stronger ones. In this case, the fisheries sector is often expected to adapt its practices in response to the expansion of OWE, particularly within the broader context of the green energy transition. Even when technical solutions are available, they may be perceived as imposed rather than negotiated, thereby limiting their legitimacy from the perspective of fishers. This is further reinforced by the cultural dimension of the conflict. Fishing can be understood as more than an economic activity, as it is often tied to identity, tradition, and multi-generational connections to specific areas. Expectations to adapt or relocate may

therefore be experienced by fishers as a loss that goes beyond material impacts, which can contribute to resistance even when solutions are technically viable.

The role of MSP becomes central in navigating these trade-offs. As outlined in the theoretical framework, MSP frames co-existence as a flexible and context-dependent arrangement rather than a guaranteed outcome, as it can facilitate but not ensure co-existence, which depends on context and actor interests. Where uses are deemed incompatible, prioritisation becomes necessary. Co-existence is often framed as possible only under certain conditions, and in some cases, not at all. This highlights a fundamental tension between co-existence and co-creation. While co-existence may be achieved through spatial or technical adjustments, co-creation would require deeper collaboration where stakeholders jointly shape solutions. The results suggest that current approaches more often align with managing co-existence than with fully realising co-creation.

This answers RQ2 by showing that co-existence is not a universal solution, but a negotiated and context-specific outcome shaped by technology, governance arrangements, knowledge systems, conflicting handling mechanisms and cultural values. It involves trade-offs where not all interests can be fully accommodated, and where adaptation is unevenly distributed across sectors. The next section discusses the role of LLs as a potential space for navigating these tensions, and examines their possibilities and limitations in supporting more collaborative forms of governance.

5.3. The Potential & Limits of LivingLabs

Comparing the perceptions of LLs and dialogue processes across the three actor groups reveals broad similarities, but also some key differences. LLs are recognised as valuable spaces for dialogue, knowledge exchange, and relationship-building across all sectors. Actors from fisheries, wind, and governance all highlight the importance of "being in the same room" and gaining insight into each other's perspectives. This suggests that LLs contribute to improved understanding and may help reduce some interpersonal tensions. Despite this shared appreciation, all sectors also point to significant limitations, particularly regarding timing, concreteness, and actual influence on decision-making processes.

The fisheries sector tends to emphasise that dialogue often occurs too late, when key decisions (such as site selection) have already been made. As a result, LLW risks being

perceived as detached from the practical realities in which fishers operate. Wind actors also acknowledge the importance of dialogue but highlight how it is often constrained by ongoing permitting processes and strategic behaviour, which can limit the willingness to share information and hinder trust-building among other involved actors. Governance actors, while generally positive, mention that LLs must be more clearly connected to formal decision-making processes to be meaningful. A quote from one of the interviewees further reinforces this limitation:

> *"Although multiple dialogue meetings have taken place between actors, they have not resulted in concrete outcomes. This indicates that dialogue alone does not necessarily translate into resolution or action." (PS3)*

While approaches, such as participatory governance, are often promoted as inclusive and legitimacy-enhancing, research (Skerritt, 2024; pp. 6) shows that they can in practice reproduce existing power imbalances. Limited transparency, unequal access to information, and differences in resources between actors may instead reinforce distrust, particularly among stakeholders who already feel marginalised. This is reflected in the fisheries sector's scepticism towards both governance processes and proposed solutions, suggesting that participation without influence risks being perceived as symbolic rather than meaningful. The findings also suggest that the purpose of dialogue processes is not necessarily to reach consensus or solutions that satisfy all stakeholders. In situations where interests compete and marine space is limited, trade-offs are often unavoidable. The value of participatory processes may therefore lie less in creating agreement and more in ensuring fair and transparent procedures where different perspectives can be heard and disagreements can be managed. In this sense, the focus shifts from achieving consensus to creating legitimate and inclusive decision-making processes.

This dynamic is further supported by Neddersen et al., (2025; pp. 12-13), who demonstrate how participatory governance in marine contexts often operates within hierarchical structures where authority remains centralised. In such systems, participation is typically limited to lower levels of involvement, such as information-sharing or consultation, while decision-making power remains with state actors. This resonates with the empirical findings of this study, where stakeholders are invited to dialogue but rarely perceive that they can shape final outcomes.

Wilcox's (1994) framework of participation provides a useful lens for interpreting these dynamics. Based on the results, LLs in this context appear to operate primarily at the levels of information and consultation, and only occasionally approach deciding together. While stakeholders are given opportunities to express their views, there is limited evidence that they are consistently involved as co-decision-makers or partners in implementation. This can be due to the actors still being rooted in old conflict, not sharing valuable information due to competing companies being present or the lack of trust in the process of participatory governance.

This gap between participation and influence helps explain why dialogue processes, despite being appreciated, are sometimes experienced as insufficient or disconnected from actual decision-making. While participatory approaches are essential for addressing complex marine conflicts, they must move beyond consultation towards more substantive forms of co-creation if they are to meaningfully support sustainable and inclusive marine governance. At the same time, deeper co-creation processes could enable stakeholders to jointly develop collective suggestions for measures and activities, which may feed into top-down decision-making and thereby create an indirect pathway for enhanced co-existence.

At the same time, LLs should not be dismissed as ineffective. They play an important role in creating access and civic standing, as described in the Trinity of Voice framework (Senecah, 2023). By bringing actors together, LLs can foster mutual understanding, strengthen trust over time, and create the conditions necessary for meaningful collaboration and innovation. However, while informal influence and even collective lobbying may occur outside formal structures, without clearer links to formal governance structures and decision-making authority, their ability to influence outcomes remains limited, and influence on outcomes remains challenging.

This answers RQ3 by showing that LLs have significant potential as platforms for dialogue and learning, but their effectiveness depends on when they are introduced, how they are designed, and whether they are connected to actual decision-making processes. In their current form, LLs within C2B2 risk functioning more as spaces for discussion than as mechanisms for redistributing influence. This is partly because a time lag between co-production activities and the realisation of impacts is to be expected in participatory processes.

5.4. Future Research

Future research could enhance this study by broadening stakeholder representation in the data collection and methods. Including recreational fisheries, more municipalities, County Administrative Boards, and potentially the defence sector would offer a more complete governance perspective. Face-to-face interviews could enrich data through non-verbal cues, though recording may still limit openness on sensitive topics. As this study reflects stakeholder perceptions rather than objective conditions, future work could integrate qualitative and quantitative data and consider multiple interpretations to strengthen validity of this study.

Additionally, future research could further explore how co-creation leads to improved co-existence over time. This includes looking more closely at different forms of power, not only limiting forms (power-over), but also enabling ones (power-to and power-with). It could also examine what different actors are able to do in practice, and how both formal structures and informal processes shape their ability to influence change towards better co-existence.

6. Conclusions & Final Remarks

This study explored how stakeholders perceive conflicts, possibilities for co-existence, and the role of dialogue processes between fisheries and OWE in the southern Skagerrak. By addressing the three RQs, the study shows that co-existence is not only a spatial issue, but one shaped by governance structures, power relations, and competing knowledge systems.

First, regarding conflicts and barriers, the findings demonstrate that tensions are deeply rooted in both spatial displacement and procedural limitations. The fisheries sector emphasizes loss of access and late inclusion in planning, while the wind sector highlights fragmented governance and unclear decision-making frameworks. Public sector actors reinforce that weak coordination and uncertainty intensify these conflicts. Importantly, the persistence of these tensions indicates that many conflicts are not new, but rather embedded in historical relationships and unresolved issues.

Second, in relation to solutions and co-existence, the study shows that co-existence is context dependent rather than generally applicable. It depends on factors such as technology, location, and fishing practices. However, a key finding is that pre-existing conflicts between

stakeholders often hinder solution-oriented discussions. This suggests that the current level of tension limits solution-oriented thinking, and that the lack of concrete solutions is itself an important result.

Third, concerning the role of LLs, the findings indicate that while they are not a solution for resolving long-standing conflicts, they can function as important arenas for offshore wind and fisheries actors to engage in for dialogue, learning, and the development of new ideas and innovation. LLs create opportunities for co-creation and may help steer processes away from the emergence of new conflicts, but their effectiveness depends on timing and their connection to actual decision-making processes.

While LLs may not always lead to immediate solutions or agreement between stakeholders, their value should not be measured solely by their ability to resolve conflicts. In situations where competing interests and trade-offs are unavoidable, LLs can contribute by creating opportunities for dialogue, increasing mutual understanding, and supporting more inclusive discussions about difficult decisions. Their greatest contribution may therefore lie not in eliminating conflict, but in creating conditions for more legitimate and inclusive decision-making about the future use of marine space.

The main contribution of this study is to highlight that sustainable co-existence requires more than technical adaptation. It requires participatory processes that are introduced earlier, allow for real influence, and acknowledge both historical tensions and different knowledge systems. To conclude, and to return to the quote by Malmberg et al., (2017; pp. 37) introduced at the beginning of this study, the findings reinforce that:

"The Living Lab, be it defined as a methodology, ecosystem or community, is foremost about bringing people together to innovate."

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Wilcox, D. (1994). THE GUIDE TO EFFECTIVE PARTICIPATION

Appendix A: Interview Guide (Swedish & English)

A1. Interview Guide (Swedish)

Sektor: Fisket **Intervjuperson #X:** Förnamn Efternamn - Organisation **Fyllt i GDPR-formulär:** JA □ | NEJ □		**Intervjuare:** Förnamn Efternamn **Intervju format:** Fysiskt □ | Online □ **Datum / tidpunkt:** ÅÅÅÅ-MM-DD, TT:MM - TT:MM
Intervjufrågor	**Anteckningar** *(följdfrågor, tankar, idéer, etc.)*	**Svar / huvudpunkter**
Bakgrund och roll		
1. Kan du kort beskriva din roll inom fisket och din erfarenhet av området i södra Skagerrak? 2. Vad för typ av fiske bedriver du?		
RQ1. Vilka konflikter och hinder identifierar intressenter mellan fisket och utbyggnaden av havsbaserad vindkraft i södra Skagerrak? *Övergripande tema: Rumsliga konflikter och hinder.*		
1. Hur upplever du planerna på havsbaserad vindkraft (HBV) i relation till ditt fiske? 2. Vilka konkreta konflikter eller problem ser du mellan fisket och HBV? 3. Finns det någon oro kring långsiktiga negativa effekter på fisket på grund av HBV?		
RQ2. Vilka förutsättningar identifierar intressenter för att möjliggöra samexistens mellan fisket och havsbaserad vindkraft? *Övergripande tema: Förutsättningar och vägar till samexistens.*		

1. Vad skulle behövas för att fisket och HBV ska kunna samexistera? 2. Skulle man kunna fiska inom samma område som HBV verkar i? 3. Har du exempel på dialoger eller samarbetsformer som fungerat bra/mindre bra sedan tidigare?		
RQ3. Hur stödjer LivingLab metodiken lärande, deltagande och gemensam förståelse i denna kontext? *Övergripande tema: Dialogprocesser och betydelsen av LLs.*		
1. Hur upplever du att fiskets perspektiv tas tillvara i planeringsprocesser idag? 2. Vad tycker du om arbetssätt som LLs eller gemensamma workshops? Vad främjar eller hindrar möjlighet för ditt deltagande? 3. Har ditt sätt att se på andra aktörer förändrats genom dialog inom LLW? 4. Har processen bidragit till ökad förståelse mellan aktörerna? Bör man fortsätta med LLs framöver?		

Sektor: Vind **Intervjuperson #X:** Förnamn Efternamn - Organisation **Fyllt i GDPR-formulär:** JA □ | NEJ □		**Intervjuare:** Förnamn Efternamn **Intervju format:** Fysiskt □ | Online □ **Datum / tidpunkt:** ÅÅÅÅ-MM-DD / TT:MM - TT:MM
Intervjufrågor	**Anteckningar** *(följdfrågor, tankar, idéer, etc.)*	**Svar / huvudpunkter**
Bakgrund och roll		
1. Kan du kort beskriva din roll i vindkraftsprojekt och er verksamhet i södra Skagerrak?		
RQ1. Vilka konflikter och hinder identifierar intressenter mellan fisket och utbyggnaden av havsbaserad vindkraft i södra Skagerrak? *Övergripande tema: Rumsliga konflikter och hinder.*		
1. Vilka utmaningar ser du i relationen mellan havsbaserad vindkraft (HBV) och fisket? 2. Finns det specifika konflikter kopplade till platsval, tillstånd eller samexistens? 3. Hur hanterar ni osäkerhet kring ekologiska effekter? 4. Finns det någon oro kring långsiktiga negativa effekter på parkerna?		
RQ2. Vilka förutsättningar identifierar intressenter för att möjliggöra samexistens mellan fisket och havsbaserad vindkraft? *Övergripande tema: Förutsättningar och vägar till samexistens.*		
1. Vilka lösningar ser du som realistiska för att möjliggöra samexistens med fisket? 2. Finns det tekniska eller designmässiga anpassningar du tror på? 3. Kan man anpassa parken så att fisket kan verka där samtidigt?		

4. Hur ser du på kompensation, samnyttjande eller gemensam planering?		
RQ3. Hur stödjer LivingLab metodiken lärande, deltagande och gemensam förståelse i denna kontext? *Övergripande tema: Dialogprocesser och betydelsen av LLs*		
1. Hur upplever du att energibolagens perspektiv tas tillvara i planeringsprocesser idag? 2. Vad tycker du om arbetssätt som LLs eller gemensamma workshops? Vad främjar eller hindrar möjlighet för ditt deltagande? 3. Har ditt sätt att se på andra aktörer förändrats genom dialog inom LLW? 4. Har processen bidragit till ökad förståelse mellan aktörerna? Bör man fortsätta med LLs framöver?		

Sektor: Offentlig sektor **Intervjuperson #X:** Förnamn Efternamn - Organisation **Fyllt i GDPR-formulär:** JA □ | NEJ □		**Intervjuare:** Förnamn Efternamn **Intervju format:** Fysiskt □ | Online □ **Datum / tidpunkt:** ÅÅÅÅ-MM-DD / TT:MM - TT:MM
Intervjufrågor	**Anteckningar** *(följdfrågor, tankar, idéer, etc.)*	**Svar / huvudpunkter**
Bakgrund och roll		
1. Kan du beskriva kommunens roll i frågor som rör havsplanering, vindkraft och fiske? 2. Hur arbetar du med avvägningar mellan olika intressen till havs?		
RQ1. Vilka konflikter och hinder identifierar intressenter mellan fisket och utbyggnaden av havsbaserad vindkraft i södra Skagerrak? *Övergripande tema: Rumsliga konflikter och hinder.*		
1. Vilka konflikter eller spänningar ser du mellan fisket och havsbaserad vindkraft? 2. Var upplever du att begränsningarna till lösningar finns idag?		
RQ2. Vilka förutsättningar identifierar intressenter för att möjliggöra samexistens mellan fisket och havsbaserad vindkraft? *Övergripande tema: Förutsättningar och vägar till samexistens.*		
1. Hur kan kommuner bidra till bättre samexistens mellan näringar till havs? 2. Finns det behov av nya planeringsverktyg eller samverkansformer? 3. Vilka lösningar ser ni för samexistens mellan fisket och havsbaserad vindkraft?		
RQ3. Hur stödjer LivingLab metodiken lärande, deltagande och gemensam förståelse i denna kontext? *Övergripande tema: Dialogprocesser och betydelsen av LLs*		

1. Vad tycker du om arbetssätt som LLs eller gemensamma workshops? Vad främjar eller hindrar möjlighet för ditt deltagande? 2. Vad krävs för att LL ska påverka beslut i praktiken? 3. Har ditt sätt att se på andra aktörer förändrats genom dialog inom LLW? 4. Har processen bidragit till ökad förståelse mellan aktörerna? Bör man fortsätta med LLs framöver?		

A2. Interview Guide (English)

Sector: Fishery **Interviewee #X:** First Name, Last Name - Organization **Signed GDPR-form:** YES □ | NO □		**Interviewer:** First Name, Last Name **Interview format:** On-site □ | Online □ **Date / Time:** YYYY-MM-DD / HH:MM - HH:MM
Interview Questions	**Notes** *(follow-up questions, thoughts, ideas, etc.)*	**Answers / key takeaways**
Background and Role		
1. Can you briefly describe your role within the fishing sector and your experience in the southern Skagerrak? 2. What type of fishing do you practice?		
RQ1. What conflicts and barriers do stakeholders identify between benthic fisheries and offshore-wind development in southern Skagerrak? *Overarching Theme: Spatial conflicts and barriers.*		
1. How do you perceive plans for OWE in relation to your fishing activities? 2. What concrete conflicts or problems do you see between fisheries and OWE? 3. Are there any concerns about long-term negative impacts on fisheries due to OWE?		
RQ2. What conditions and pathways do stakeholders identify for enabling co-existence between fisheries and OWE? *Overarching Theme: Conditions and pathways for co-existence.*		
1. What would be required for fisheries and OWE to co-exist? 2. Would it be possible to fish within the same areas where OWE operates?		

3. Do you have examples of dialogue or forms of collaboration that have worked well or less well in the past?		
RQ3. How does the LivingLab method support learning, participation and shared understanding in this context? *Overarching Theme: Dialogue processes and the role of LLs.*		
1. How do you perceive that the perspectives of the fishing sector are taken into account in planning processes today? 2. What do you think about approaches such as LLs or joint workshops? What enables or limits your ability to participate? 3. Has your perception of other actors changed through dialogue within LLW? 4. Has the process contributed to increased understanding between actors? Do you think LLs should be continued in the future?		

Sector: Wind **Interviewee #X:** First Name, Last Name - Organization **Signed GDPR-form:** YES □ | NO □		**Interviewer:** First Name, Last Name **Interview format:** On-site □ | Online □ **Date / Time:** YYYY-MM-DD / HH:MM - HH:MM
Interview Questions	**Notes** *(follow-up questions, thoughts, ideas, etc.)*	**Answers / key takeaways**
Background and Role		
1. Can you briefly describe your role in offshore wind projects and your operations in the southern Skagerrak?		
RQ1. What conflicts and barriers do stakeholders identify between benthic fisheries and offshore-wind development in southern Skagerrak? *Overarching Theme: Spatial conflicts and barriers.*		
1. What challenges do you see in the relationship between OWE and fisheries? 2. Are there specific conflicts related to site selection, permitting, or co-existence? 3. How do you handle uncertainty regarding ecological impacts? 4. Are there any concerns about long-term negative impacts on offshore wind farms?		
RQ2. What conditions and pathways do stakeholders identify for enabling co-existence between fisheries and OWE? *Overarching Theme: Conditions and pathways for co-existence.*		
1. What solutions do you see as realistic for enabling co-existence with fisheries? 2. Are there technical or design adaptations that you consider promising? 3. Is it possible to adapt wind farms so that fishing can take place within the same areas?		

4. How do you view compensation, co-use, or joint planning?		
RQ3. How does the living lab method support learning, participation and shared understanding in this context? *Overarching Theme: Dialogue processes and the role of LLs.*		
1. How do you perceive that the perspectives of the wind sector are taken into account in planning processes today? 2. What do you think about approaches such as LLs or joint workshops? What enables or limits your ability to participate? 3. Has your perception of other actors changed through dialogue within LLW? 4. Has the process contributed to increased understanding between actors? Do you think LLs should be continued in the future?		

Sector: Public Sector **Interviewee #X:** First Name, Last Name - Organization **Signed GDPR-form:** YES □ | NO □		**Interviewer:** First Name, Last Name **Interview format:** On-site □ | Online □ **Date / Time:** YYYY-MM-DD / HH:MM - HH:MM
Interview Questions	**Notes** *(follow-up questions, thoughts, ideas, etc.)*	**Answers / key takeaways**
Background and Role		
1. Can you describe the municipality's role in issues related to marine spatial planning, offshore wind energy, and fisheries? 2. How do you work with balancing different interests at sea?		
RQ1. What conflicts and barriers do stakeholders identify between benthic fisheries and offshore-wind development in southern Skagerrak? *Overarching Theme: Spatial conflicts and barriers.*		
1. What conflicts or tensions do you see between fisheries and offshore wind energy (OWE)? 2. Where do you perceive the main limitations to solutions today?		
RQ2. What conditions and pathways do stakeholders identify for enabling co-existence between fisheries and OWE? *Overarching Theme: Conditions and pathways for co-existence.*		
1. How can municipalities contribute to better co-existence between different marine sectors? 2. Is there a need for new planning tools or forms of collaboration? 3. What solutions do you see for co-existence between fisheries and offshore wind energy?		

RQ3. How does the LivingLab method support learning, participation and shared understanding in this context? *Overarching Theme: Dialogue processes and the role of LLs.*		
1. What do you think about approaches such as LLs or joint workshops? What enables or limits your ability to participate? 2. What is required for LLs to influence decision-making in practice? 3. Has your perception of other actors changed through dialogue within LLW? 4. Has the process contributed to increased understanding between actors? Do you think LLs should be continued in the future?		